AMANA

Secure Device Registration and
Second-Hand Transaction Safety Platform

SOFTWARE REQUIREMENTS SPECIFICATION

Version 4.0 · Consolidated implementation baseline · 20 October 2026

Document field	Value
Document status	Draft for supervisor review and approval
Project type	Academic cybersecurity and full-stack web application
Implementation stack	PostgreSQL, Express.js, React, Node.js (PERN)
Primary users	Device claimants, second-hand buyers, authorized reviewers, senior reviewers, system administrators
Languages	English and Amharic
Requirements basis	ISO/IEC/IEEE 29148 requirements-engineering principles; not a claim of formal certification

Purpose: define a privacy-conscious, auditable way to register device claims, review theft reports, manage disputes, and complete eligible AMANA-approved transfers without claiming to prove legal ownership.

Prepared as a consolidated specification. Policy-sensitive retention periods and project authority appointments require supervisor approval before production use.

Document control

Version	Date	Summary	Status
3.0	Earlier baseline	Separated report, investigation and restriction states; initial case/account rules.	Superseded
3.1	Earlier baseline	Added permission matrix, identifiers, recipient acceptance and dispute workflow.	Superseded
3.2	Earlier baseline	Expanded operational rules, database fields, security controls and tests.	Superseded
4.0	20 Oct 2026	Consolidated formal structure and detailed workflows; resolved major rule conflicts and added decision/traceability controls.	Current draft

Approval and use

This SRS is the implementation baseline only after supervisor approval. Any policy change must be recorded as a versioned change, reviewed against affected requirements and tests, and approved before implementation.

Contents

- 1. Introduction and scope
- 2. Definitions and state model
- 3. Core use cases and scenarios
- 4. Business rules
- 5. Permissions and effective status
- 6. Theft report and investigation workflow
- 7. Identifier conflict workflow
- 8. Claim-dispute workflow
- 9. Evidence and receipt review
- 10. AMANA-approved transfer workflow
- 11. Functional requirements
- 12. Data requirements and integrity
- 13. Security, privacy and abuse controls
- 14. Notifications and UI
- 15. Non-functional requirements
- 16. Acceptance tests and traceability
- 17. Deployment and operations
- 18. Retention and unresolved approvals
- 19. Definition of done
- 20. Approval block and change log

1. Introduction and scope

1.1 Purpose

AMANA helps people make more informed decisions about second-hand phones and computers by recording device identifiers, user claims, supporting evidence, theft allegations, investigation decisions, restrictions, and AMANA-approved transfer history. It provides a limited public status lookup and controlled authenticated workflows.

1.2 Product objectives

- Allow a claimant to register a phone or computer using a supported identifier.
- Accept optional purchase-receipt evidence and permit later submission against the same claim.
- Distinguish a theft allegation, an active investigation, and a substantiated finding.
- Prevent eligible transfers from bypassing current restrictions, competing claims, or recipient consent.
- Retain proportionate audit history for dispute and abuse review while protecting personal information.
- Give public lookup users a cautious status result without exposing claimant or reporter identities.

1.3 In scope

Account registration, authentication, password recovery and deactivation; device and identifier registration; receipt submission and review; theft reporting and investigation; claim disputes; identifier-conflict review; association release; AMANA-approved transfer; public lookup; audit history; notifications; rate limiting; English and Amharic UI; and security tests.

1.4 Out of scope

GPS tracking, remote locking, device recovery, payments, a marketplace, blockchain, AI-based fraud detection, automated vulnerability scanning, and integrations with police, Fayda, telecom operators, manufacturers, or government systems. AMANA does not guarantee legal ownership, genuine hardware identity, or that a device has never been stolen.

1.5 Standards note

The document follows requirements-engineering practices associated with ISO/IEC/IEEE 29148. It does not claim formal certification or legal compliance. Applicable privacy and legal obligations must be reviewed before production deployment.

2. Definitions and state model

Term	Normative definition
Device	A physical phone or computer represented by one internal AMANA device record.
Device identifier	An IMEI or manufacturer-assigned computer serial number associated with a device record.
Claim	A user's recorded assertion of association with a device; it is not by itself proof of legal ownership.
Claimant	The user linked to a device through an active claim/association.
Affected claimant	The claimant whose association or permissions are directly affected by a case. A reporter is not automatically the affected claimant.
Receipt status	One of receipt-free, receipt-pending, receipt-accepted, or receipt-rejected.
Acquisition basis	The reason the claim was created, stored separately from receipt status: accepted receipt-backed claim, completed AMANA-approved transfer, or other/unverified basis.
Theft report	An allegation that a device may be stolen.
Investigation	A formal case created after a report passes screening.
Provisional restriction	A temporary restriction applied when a report passes screening, subject to reviewer triage within 72 hours.
Active theft restriction	A restriction supported by a documented substantiated finding.
Inconclusive restriction	A restriction retained after an inconclusive finding with a recorded reason and review date.
Claim-dispute case	A process to assess competing claims without creating a second active association.
Identifier-conflict case	A review of a newly submitted identifier that conflicts with an existing record; the new submission is held, not the existing claimant by default.
Association release	Ending a claimant's active association. It does not prove sale, transfer legal title, delete the device, or erase history.
AMANA-approved transfer	A transfer completed through the authenticated workflow after eligibility checks and recipient acceptance.
Account deactivation	Disabling ordinary login/use while retaining records required for active cases, security, audit, and approved retention rules.
Reviewer / senior reviewer	Authorized case decision-maker / reviewer permitted to approve extensions and independent reconsiderations.
System administrator	Operator of technical configuration and access. Administrative access alone does not authorize case decisions.

2.1 Separate state dimensions

The system shall not collapse these independent dimensions into one status: (1) receipt evidence status, (2) acquisition basis, (3) claim status, (4) association status, (5) report/investigation status, (6) restriction status, (7) transfer status, and (8) account status.

2.2 Effective status

The device's effective public status is derived at request time from current restrictions and cases using the precedence in Section 5. It is not independently editable. Historical reports alone do not create a current restriction.

2.3 Normative wording

“Shall” denotes a mandatory requirement. “May” denotes permitted behavior. “Should” denotes a recommendation that requires a documented exception if not followed.

3. Core use cases and scenarios

UC-01: Register a device

A registered user submits an IMEI or computer serial number and creates a claim. Receipt evidence is optional. The system validates syntax and uniqueness, checks current restrictions, records the acquisition basis and receipt status separately, and does not represent registration as proof of ownership.

UC-02: Submit and review receipt evidence

A claimant uploads a supported receipt file during registration or later. Evidence remains pending until an authorized reviewer accepts, rejects, or requests correction with a recorded reason. Acceptance does not remove unrelated restrictions or authorize past transfers.

UC-03: Report possible theft

An authenticated user submits a report. Screening checks required fields, identifier syntax, duplicate patterns and rate limits. A passing report atomically creates or links an investigation and applies a provisional restriction. A reviewer triages within 72 hours.

UC-04: Resolve competing claims

A second claimant cannot create a second active association. The request is held and routed to a claim-dispute case. The current claimant may voluntarily release their association, but release does not clear independent restrictions or prove the new claimant's ownership.

UC-05: Transfer a device

An eligible claimant initiates a transfer to a recipient. The recipient must explicitly accept within 72 hours. The server rechecks eligibility and restrictions immediately before transactional completion. Decline, cancellation or expiry leaves the source association unchanged.

UC-06: Look up a device

A user enters a supported identifier and receives only a generic effective status and explanation. The result must not reveal whether a named person registered the device, who reported it, or what evidence reviewers saw.

Scenario 1: Receipt accepted, no current blocker

The claim may initiate an AMANA-approved transfer if its acquisition basis is eligible, the association is active, no restriction or dispute blocks the operation, and the recipient accepts. Association release is also allowed when no hold applies.

Scenario 2: No accepted receipt, no current blocker

A claim with no accepted receipt and no completed AMANA-transfer basis cannot initiate an AMANA-approved transfer. The claimant may release the association if no blocking condition applies. Release is not a verified sale or transfer. A later claimant must create a new claim and cannot inherit transfer eligibility merely because the prior claimant released.

Scenario 3: Theft report and receipt-backed claim

A passing report creates or joins the active investigation and applies a provisional restriction. Triage is due within 72 hours. Investigation deadline: 720 hours from opening, UTC. One extension of up to 360 hours may be approved, with an absolute cap of 1,080 hours from opening.

Scenario 4: Theft report and no accepted receipt, or no current claimant

The same deadlines and review controls apply. The case records the evidence/acquisition state at opening for audit and decision context. Later evidence is considered but does not silently reset the deadline.

4. Business rules

ID	Rule
BR-001	A historical report alone does not block an operation. Current restrictions and active cases govern permissions.
BR-002	Claim evidence status, acquisition basis, association status, case status, restriction status, transfer status and account status shall remain separate.
BR-003	Receipt acceptance requires reviewer assessment; it is never automatic.
BR-004	A claim is eligible for transfer only when its acquisition basis is accepted receipt-backed claim or completed AMANA-approved transfer, and all other checks pass.
BR-005	A claim without an eligible acquisition basis cannot initiate an AMANA-approved transfer. Later evidence may be submitted against the same claim.
BR-006	Association release is not a transfer, does not delete the device or history, and does not prove a private sale.
BR-007	At most one active claimant association may exist per device.
BR-008	Every new theft investigation has a 720-hour deadline measured from its opening timestamp in UTC.
BR-009	One extension of at most 360 hours may be approved by a senior reviewer with a recorded reason. The total cap is 1,080 hours from opening.
BR-010	Cases approaching the cap shall be escalated. At the cap, a designated case authority must take ownership and record a decision or a specific next action and review date; no automatic theft finding is allowed.
BR-011	A provisional restriction must be triaged by an authorized reviewer within 72 hours: confirm continuation, dismiss a clearly baseless report with reason, or escalate. Failure is escalated to the senior queue.
BR-012	Only one active theft investigation per device is permitted. Additional reports link to it and do not reset the deadline.
BR-013	A closed case may be reopened only after materially new information is accepted for review; prior decisions remain in history.
BR-014	Reconsideration reviews an existing decision. Reopening formally reactivates a closed case based on materially new information.
BR-015	An inconclusive outcome may retain a restriction only with documented necessity, a named accountable reviewer, a review date no more than 30 days away, and a reconsideration route.
BR-016	A user may report their own device. The user retains case access and communication rights; account deactivation is blocked while they are the affected claimant in an active investigation.
BR-017	Reporting another person's device alone does not block the reporter's account deactivation.
BR-018	A pending transfer expires after 72 hours unless completed; the initiator may cancel earlier. Cancellation/expiry releases only that transfer's hold.
BR-019	An active claim dispute blocks AMANA-approved transfer but permits the current claimant to voluntarily release their association. Independent restrictions remain.
BR-020	A new conflicting identifier submission is held for review. The existing claimant is not restricted solely because of that submission.
BR-021	Account deactivation is blocked while the user has an affected active investigation, active claim dispute involving their association, a pending transfer they initiated, or a pending reconsideration request.
BR-022	If no blocking condition exists, deactivation ends an unrestricted active association and retains its historical record.
BR-023	After a case closes, a user may deactivate even if a restriction remains. The restricted association is retained but must have a defined resolution path, accountable authority, review date, and appeal/reconsideration route.
BR-024	If a restricted device has no active claimant, a new user may submit a claim for review; registration does not clear the restriction or permit transfer/release.
BR-025	A transfer requires explicit recipient acceptance and a final server-side eligibility check.
BR-026	Public lookup must not reveal whether a specific person has registered or claimed a device, or disclose identities, contact details, private evidence or reviewer notes.
BR-027	Material claim, association, report, case, restriction, identifier, transfer, reviewer, reconsideration and deactivation changes shall be audit logged.
BR-028	A reviewer shall not decide a case involving their own device, claim, report, close relationship or declared personal interest. System checks plus reviewer self-declaration are required.
BR-029	An unresolved identifier-conflict or dispute hold applies to the new submission/disputed transfer as defined in the relevant workflow; it shall not silently freeze an existing claimant.
BR-030	Public lookup results are informational and are not a legal ownership certificate or guarantee that no theft report exists outside AMANA.

5. Permissions and effective public status

5.1 Closed list of blocking conditions

Only the following recorded conditions may block operations: (1) active theft investigation with provisional restriction; (2) active theft restriction from a substantiated finding; (3) retained inconclusive restriction; (4) active claim dispute, transfer-blocking only while allowing voluntary release; (5) unresolved identifier conflict affecting the new submission; and (6) pending transfer, which blocks release/deactivation for the initiating source association until cancellation, expiry or completion. No generic unrecorded “other blocking condition” is permitted.

5.2 Permission matrix

Condition	Transfer	Release	Deactivate	New claim	Public label
No blocker; eligible basis	Allowed after recipient acceptance	Allowed	Allowed; ends unrestricted association	Only if no active association	No active theft restriction found
No blocker; ineligible basis	Blocked	Allowed	Allowed if no other hold	Only if no active association	No active theft restriction found
Receipt pending/rejected; no other blocker	Blocked	Allowed	Allowed if no active case/transfer	Existing claim may receive later evidence	No active theft restriction found
Active theft investigation	Blocked	Blocked for affected association	Blocked for affected claimant	If no claimant, may record restricted claim	Under review
Active theft restriction	Blocked	Blocked until authorized lift/reassignment	Allowed after case closure; restricted association retained	May submit for review if no active association; remains restricted	Active theft restriction
Inconclusive restriction retained	Blocked	Blocked until authorized lift/reassignment	Allowed after case closure; association retained	May submit for review if no active association; remains restricted	Restricted pending further review
Active claim dispute	Blocked	Allowed; ends association dispute only	Blocked while dispute active unless association released and no other hold	Linked to dispute; no second active association	Restricted pending review
New identifier conflict	Blocked for new conflicting submission	Existing claimant may release if no other hold	Existing claimant may deactivate if no other hold	New submission held	Restricted pending review
Pending transfer	Pending until accepted and checks pass	Blocked for source association until cancel/expiry	Blocked for initiator until cancel/expiry	No second active association	Based on independent current status
Deactivated, unrestricted association ended	Unavailable to inactive account	Already ended	Already deactivated	Allowed only if no active association/restriction	Based on current restrictions
Deactivated, restricted association retained	Blocked	Blocked until authorized resolution	Already deactivated	May submit for review but cannot replace retained association	Based on current restriction

5.3 Public status precedence

The effective public status shall use this order: (1) Under review for an active theft investigation; (2) Active theft restriction; (3) Restricted pending further review for a retained inconclusive restriction; (4) Restricted pending review for dispute or identifier-conflict restrictions; (5) No active theft restriction found. If more than one condition applies, the highest-precedence label is returned. No public label may reveal claimant identity or whether a named person has registered the device.

5.4 Server-side enforcement

Every protected operation must re-evaluate current account, association, acquisition-basis, transfer, case and restriction state on the server at execution time. UI state is not authoritative. State-changing operations must be transactionally protected against concurrent requests.

6. Theft report and investigation workflow

6.1 Minimum report fields

Required: authenticated reporter; identifier type and value; allegation category; description; accuracy declaration. Incident date/period is required if known and otherwise may be omitted with a reason. Supporting evidence is optional. A receipt is not mandatory to report suspected theft.

6.2 Screening and triage

Screening checks authentication, required fields, identifier syntax, active case duplication, materially identical reports without new information, and rate limits. Missing required fields or invalid identifier syntax returns the submission for correction and does not activate a restriction. A duplicate report links to the existing case. A passing report atomically creates or links the active case, applies a provisional restriction, and records the audit event. Reviewer triage is due within 72 hours.

6.3 Deadline and escalation

- Initial deadline: 720 hours from case opening, UTC.
- One extension: up to 360 hours, approved by a senior reviewer with reason and audit record.
- Absolute cap: 1,080 hours from the original opening timestamp; extension does not reset the cap.
- Warning: 72 hours before the current deadline.
- At the cap: escalate to senior reviewer and designated project case authority. No automatic outcome.

If no final decision exists at the cap, the designated authority must accept assignment, record why the case remains unresolved, specify the next review action, and set a review date no more than 30 days away. Every continuing restriction must have a named accountable authority and review date. This is an exception-controlled continuing review, not an automatic finding of theft.

6.4 Decision criteria

A finding must be based on the totality of documented evidence, identifier consistency, chronology, source reliability, material contradictions, and the claimant/reporter's opportunity to respond. A receipt is relevant but not conclusive by itself. Reviewers must distinguish:

- **Substantiated:** documented evidence provides a reasonable, recorded basis to conclude the theft allegation is supported under the project's approved review standard.
- **Unsubstantiated:** evidence does not sufficiently support the allegation.
- **Invalid:** the report is malformed, materially false on verifiable facts, outside scope, or otherwise ineligible; record a specific reason. An unsubstantiated report is not automatically malicious.
- **Inconclusive:** available evidence does not support a reliable determination. Restriction may be retained only under BR-015.

6.5 Outcome-to-restriction mapping

Outcome	Restriction effect	Association/permissions
Substantiated	Apply/retain active theft restriction; set next review and accountable authority.	Transfer/release blocked until authorized lift or reassignment. Account deactivation allowed after case closure, subject to other holds.
Unsubstantiated	Remove only the provisional restriction attributable to this allegation.	Normal permissions resume unless another independent blocker applies.
Invalid	Remove only the provisional restriction attributable to this report.	Normal permissions resume unless another independent blocker applies.
Inconclusive	Close investigation; retain restriction only with documented necessity and review date ≤30 days.	Transfer/release remain blocked while retained restriction is active; reconsideration available.
Reopened	Record new opening timestamp, reason and new 720-hour deadline; prior history remains.	A new provisional restriction applies only if explicitly authorized in the reopening decision.

6.6 Reconsideration and reopening

An affected user may request reconsideration within 30 calendar days of a decision, stating the decision, grounds and available supporting evidence. A senior reviewer who did not make the original decision shall decide within 10 calendar days. One reconsideration request per decision is allowed, unless materially new evidence supports a separate reopening request. Deactivation is blocked while reconsideration is pending. A deactivated user must have secure limited case access or identity recovery to submit the request. Reconsideration reviews the existing decision; reopening reactivates a closed case only after materially new information is accepted. Prior decisions and audit events remain.

6.7 Reviewer assignment and independence

New cases enter a queue and are assigned to the next available qualified reviewer using round-robin or least-open-case allocation. The system must exclude reviewers with system-detectable conflicts (their own user/device/claim/report) and require a recorded self-declaration of no other personal interest. A reviewer must recuse if a conflict emerges. Escalations go to the senior-reviewer queue; system administrators cannot decide cases solely by virtue of administrator role.

7. Identifier-conflict workflow

The system normalizes and validates the submitted identifier and checks for an existing association. A conflict holds the new submission, not the existing claimant by default. The existing claimant's transfer, release and deactivation permissions do not change solely because a new submission conflicts.

- Initial deadline: 15 calendar days (360 hours) from opening, UTC.
- One extension: up to 7 calendar days (168 hours), approved by a senior reviewer with reason.
- Maximum: 22 calendar days (528 hours). Warning 72 hours before current deadline.

The reviewer records whether the submission is a duplicate, data-entry error, legitimate additional identifier, or unresolved conflict. Confirmed invalid submissions are rejected without changing the existing record. An unresolved conflict at the deadline is escalated and remains attached only to the new submission pending an accountable next action and review date no more than 30 days away. Device records shall not be merged solely because identifiers look similar. An existing claimant may be restricted only by an independent qualifying case or documented authorized decision.

8. Claim-dispute workflow

A competing claimant submits the device identifier, acquisition basis, description and available supporting evidence. Screening checks required fields, rate limits, active claims and cases. A second active association is never created. A passing request creates a dispute case that blocks AMANA-approved transfer but permits voluntary release by the current claimant.

Voluntary release ends the association dispute because the previous active association no longer needs to be contested. It does not prove that the new claimant owns the device and does not remove an independent theft restriction or unresolved identifier conflict.

- Initial deadline: 15 calendar days (360 hours), UTC.
- One extension: up to 7 calendar days (168 hours), approved by a senior reviewer.
- Maximum: 22 calendar days (528 hours), with warning 72 hours before deadline.

If the dispute reaches its cap unresolved, it is escalated to the senior reviewer and designated case authority. The authority must record a decision or accountable next action and review date no more than 30 days away. The outcome records the basis for ending or retaining an association. A dispute decision does not establish legal ownership.

9. Evidence and receipt review

Evidence files are private by default and are accessible only to the submitting user where appropriate, assigned reviewers, authorized senior reviewers and technical roles with a documented operational need. Evidence URLs must not be public or permanently guessable.

- Supported formats: PDF, JPEG and PNG only.
- Maximum file size: 10 MB per file; maximum five files per submission.
- Validate extension, declared MIME type, file signature, size and safe generated storage name. Store outside the public web root or in private object storage.

- Receipt evidence may be submitted at registration or later against the same claim.
- Receipt review target: within five calendar days of submission; overdue items escalate to the senior queue. Pending evidence is never treated as accepted.
- Reviewer outcome: accept, reject, or request correction, with a reason and audit event.

Receipt acceptance changes evidence status through a new review event and does not retroactively authorize transfers or remove independent restrictions. A receipt does not conclusively prove legal ownership.

10. AMANA-approved transfer workflow

• Source claimant initiates a transfer; server verifies eligible acquisition basis and current permissions.

• System records a pending transfer with a 72-hour expiry and sends an in-app notification; email is sent when configured.

• Recipient must explicitly accept within 72 hours. Recipient may decline; source claimant may cancel before completion.

• On acceptance, the server rechecks the source association, acquisition basis, active restrictions, cases, account state and recipient eligibility.

• Completion changes association/claim state in one database transaction and records the recipient's acquisition basis as “Acquired via AMANA-approved transfer.”

• If declined, cancelled, or expired, source association remains unchanged and only the pending-transfer hold is released.

• Every initiation, acceptance, decline, cancellation, expiry and completion is audit logged.

An AMANA-approved transfer is a platform-recorded handover, not a payment service or a legal ownership adjudication. A recipient may later resell through AMANA if the current eligibility checks pass.

11. Functional requirements

11.1 Account management

ID	Requirement
FR-ACC-001	The system shall support account registration, authentication, logout and secure password recovery.
FR-ACC-002	The system shall enforce authorization on every protected server operation.
FR-ACC-003	The system shall support account deactivation according to the permission matrix.
FR-ACC-004	Deactivation shall not cascade-delete required device, claim, case, decision, transfer or audit records.
FR-ACC-005	Deactivation shall be blocked for the holds specified in BR-021.
FR-ACC-006	A deactivated user shall have a secure limited-access or identity-recovery route for pending case requests.

11.2 Device, identifiers and claims

ID	Requirement
FR-DEV-001	The system shall support phone IMEIs and manufacturer-assigned computer serial numbers.
FR-DEV-002	An IMEI shall contain exactly 15 decimal digits and pass a Luhn checksum before syntactic acceptance.
FR-DEV-003	A dual-SIM phone may have multiple IMEIs linked to one device record.
FR-DEV-004	Computer serial numbers shall retain identifier type and normalized representation; no universal manufacturer format shall be assumed.
FR-DEV-005	Syntax validation shall not be described as proof of genuine identity or ownership.
FR-DEV-006	Conflicting or duplicate identifiers shall follow Section 7.
FR-DEV-007	Receipt evidence shall be accepted during registration or later against an existing claim.
FR-DEV-008	Receipt status shall distinguish receipt-free, receipt-pending, receipt-accepted and receipt-rejected.
FR-DEV-009	Later receipt acceptance shall be recorded and shall not authorize earlier transfers retroactively.
FR-DEV-010	Acquisition basis shall be stored separately from receipt status.
FR-DEV-011	The system shall prevent more than one active claimant association per device at database level.

11.3 Evidence review

ID	Requirement
FR-EVD-001	Every evidence submission shall enter a reviewer queue.
FR-EVD-002	A reviewer shall accept, reject or request correction with a recorded reason.
FR-EVD-003	Receipt review shall be completed within five calendar days or escalated.
FR-EVD-004	Pending evidence shall not be treated as accepted.
FR-EVD-005	Evidence decisions shall not automatically lift independent restrictions.
FR-EVD-006	Evidence review history shall preserve prior states and decisions.

11.4 Reports and investigations

ID	Requirement
FR-INV-001	A passing theft report shall create an investigation or link to the active investigation for that device.
FR-INV-002	Report creation/linking, provisional restriction and audit event shall be atomic.
FR-INV-003	Every new investigation shall receive a 720-hour UTC deadline.
FR-INV-004	A reviewer shall triage provisional restrictions within 72 hours.
FR-INV-005	The system shall warn the assigned reviewer 72 hours before the current deadline.
FR-INV-006	One extension of up to 360 hours may be approved by a senior reviewer with a reason.
FR-INV-007	A case reaching 1,080 hours without final decision shall be escalated to the designated authority.
FR-INV-008	A final outcome shall be substantiated, unsubstantiated, invalid or inconclusive.
FR-INV-009	Unsubstantiated or invalid outcomes shall remove only their own provisional restriction.
FR-INV-010	An inconclusive outcome may retain a restriction only with documented reason and review date no more than 30 days away.
FR-INV-011	Reopening shall require materially new information accepted for review.
FR-INV-012	Reopening shall record a new opening time, deadline, reason and restriction decision without erasing prior history.
FR-INV-013	The system shall enforce reviewer independence, recusal and senior escalation.

11.5 Disputes

ID	Requirement
FR-DSP-001	A competing claim shall create a dispute case rather than a second active association.
FR-DSP-002	An active dispute shall block transfer but allow voluntary association release.
FR-DSP-003	Voluntary release shall end the association dispute but not independent restrictions.
FR-DSP-004	A dispute shall have a 360-hour initial deadline and at most one 168-hour extension.
FR-DSP-005	Overdue disputes shall be escalated and have an accountable next action/review date.
FR-DSP-006	Dispute outcomes and association changes shall be audit logged.

11.6 Transfers

ID	Requirement
FR-TRN-001	A transfer shall require an eligible acquisition basis and no current transfer-blocking condition.
FR-TRN-002	The recipient shall explicitly accept within 72 hours.
FR-TRN-003	The initiator may cancel before completion.
FR-TRN-004	A transfer shall expire after 72 hours without acceptance.
FR-TRN-005	The server shall recheck all eligibility and restriction conditions immediately before completion.
FR-TRN-006	A completed transfer shall record acquisition basis as “Acquired via AMANA-approved transfer.”
FR-TRN-007	Transfer completion and association changes shall be transactional.
FR-TRN-008	A recipient who acquired via AMANA may later request another transfer if all current checks pass.

11.7 Lookup

ID	Requirement
FR-LKP-001	Public lookup shall return only an effective status and generic explanatory text.
FR-LKP-002	Public lookup shall not reveal whether a named person has registered or claimed a device.
FR-LKP-003	Public lookup shall not disclose claimant/reporter identity, contact details, evidence or reviewer notes.
FR-LKP-004	Status precedence shall follow Section 5.3.
FR-LKP-005	Dispute and identifier conflict shall use the generic “Restricted pending review” label.
FR-LKP-006	“No active theft restriction found” shall not imply safety, legal ownership, or absence of unreported theft.

12. Data requirements and integrity

12.1 Minimum entities and fields

The following are minimum logical fields. Implementation may add fields but shall preserve the meanings and relationships. Use UUIDs or non-guessable internal IDs where appropriate; public endpoints must not expose sequential internal IDs.

Entity	Minimum fields
User	id; email (unique normalized); password_hash; role; account_status; created_at; deactivated_at; locale; email_verified_at
Device	id; device_type; record_status; created_at; updated_at
DeviceIdentifier	id; device_id; identifier_type; normalized_value; review_status; created_at; ended_at
DeviceClaim	id; device_id; claimant_user_id; receipt_status; acquisition_basis; claim_status; created_at; ended_at
DeviceAssociation	id; device_id; claim_id; user_id; association_status; start_at; end_at; end_reason
Evidence	id; claim_id nullable; case_id nullable; submitted_by; private_storage_key; mime_type; file_size; checksum; review_status; created_at
EvidenceReview	id; evidence_id; reviewer_id; decision; reason; created_at
TheftReport	id; device_id; reporter_user_id; allegation_category; description; incident_period nullable; screening_status; case_id; created_at
InvestigationCase	id; device_id; case_status; assigned_reviewer_id; opened_at; deadline_at; extension_deadline_at; closed_at; outcome; outcome_reason; next_review_at; accountable_authority_id
ClaimDisputeCase	id; device_id; requester_user_id; case_status; assigned_reviewer_id; opened_at; deadline_at; extension_deadline_at; closed_at; outcome; outcome_reason; next_review_at; accountable_authority_id
IdentifierConflictCase	id; device_identifier_id; submitted_by; existing_device_id; case_status; assigned_reviewer_id; opened_at; deadline_at; extension_deadline_at; outcome; outcome_reason; next_review_at
Restriction	id; device_id; restriction_type; source_case_type; source_case_id; reason; status; started_at; ended_at; next_review_at; accountable_authority_id
Transfer	id; device_id; source_claim_id; recipient_user_id; status; acquisition_basis; initiated_at; expires_at; accepted_at; completed_at; cancelled_at
ReconsiderationRequest	id; case_type; case_id; requester_user_id; reason; status; assigned_reviewer_id; submitted_at; decision_at; outcome
AuditEvent	id; actor_user_id nullable; action; target_type; target_id; case_type nullable; case_id nullable; created_at; reason; correlation_id
Notification	id; recipient_user_id; channel; type; delivery_status; created_at; sent_at; retry_count

12.2 Required relationships and constraints

- At most one active DeviceAssociation per device, enforced by a database uniqueness/partial-index strategy and transaction logic.
- At most one active InvestigationCase per device, enforced by a database constraint or equivalent serializable locking strategy.
- Identifier uniqueness is enforced after normalization; additional identifiers on one device are allowed only through the defined reviewed association flow.

- A completed transfer must atomically end the source association and establish the recipient association, preserving the one-active-association invariant.
- A transfer completion transaction must lock/recheck the device and all current blocking restrictions before committing.
- Report screening, case creation/linking, provisional restriction creation and audit event must commit or roll back together.
- Effective public status is derived from current case/restriction records and precedence; it is not an independently editable field.
- Historical claims, associations, reports, outcomes and transfers are not overwritten to erase prior states.
- Account deactivation must not cascade-delete records required for cases, security, audit or approved retention.
- Every restriction must have a source or documented authority, start time, reason, accountable owner and next review date if it continues.

12.3 Transaction and concurrency behavior

For any operation that could race (transfer completion, association release, case opening, restriction change, new claim), the backend must use database transactions and row-level locking or an equivalent isolation strategy. The database remains the final guard against duplicate active associations. Failed concurrent operations must return a safe conflict response and create an audit event where appropriate.

13. Security, privacy and abuse controls

ID	Requirement
SEC-001	Use a modern password-hashing algorithm supported by the selected stack; never store plaintext passwords.
SEC-002	Enforce server-side authorization for every protected operation and object; prevent IDOR/BOLA.
SEC-003	Use parameterized queries or safe ORM/query-builder operations.
SEC-004	Protect cookie-authenticated state-changing requests with CSRF controls and secure cookie settings.
SEC-005	Render user-controlled text safely and apply context-appropriate output encoding to prevent XSS.
SEC-006	Accept PDF, JPEG and PNG only; max 10 MB per file and five files per submission.
SEC-007	Validate file extension, MIME declaration, signature and size; use generated names and private storage outside public web root.
SEC-008	Initial rate limits: login 10 failed attempts/account/15 min; recovery 5 requests/account/hour; lookup 60/IP/min and 20/account/min; theft reports 5/user/rolling 24 h; disputes 3/user/rolling 24 h; reconsideration 3/user/rolling 24 h; evidence upload 10 attempts/user/hour.
SEC-009	Rate-limit events shall be logged and temporarily rejected; rate limits alone shall not label a user malicious.
SEC-010	Reviewer triage of provisional restrictions is due within 72 hours.
SEC-011	Repeated materially identical reports without new information shall link to an active case or be rejected as duplicates with reason.
SEC-012	An unsubstantiated report shall not automatically be classified as malicious.
SEC-013	Audit events shall be protected against ordinary user modification/deletion; privileged tampering attempts must be logged.
SEC-014	Reviewer assignment, self-declared conflicts, recusal, decisions, extensions, restriction changes and escalations shall be logged.
SEC-015	Production secrets shall not be committed to source control or exposed in client bundles.
SEC-016	Production deployment shall use HTTPS, restricted database access, protected backups, documented dependency updates and appropriate secure cookies.
SEC-017	Public responses shall not expose stack traces, secrets, private evidence or internal database details.
SEC-018	Public lookup shall resist enumeration through rate limits, uniform response structure, non-disclosing error messages and monitoring.
SEC-019	Case and evidence access shall follow least privilege; all access to private evidence shall be authorized and auditable.
SEC-020	Security logs shall avoid storing passwords, session tokens, full private evidence, or unnecessary personal data.

14. Notifications and UI

14.1 Notification channels

The initial release shall support in-app notifications and email when a valid address and delivery configuration exist. Required events: report receipt, case opening/closure, triage outcome, deadline warning, extension decision, evidence decision, dispute outcome, transfer accepted/declined/cancelled/expired/completed, and reconsideration decision. Email must contain minimal information and no private evidence attachments; the authenticated case view is authoritative.

14.2 User interface requirements

ID	Requirement
UI-001	Public lookup explains status meaning and limitations.
UI-002	Registration distinguishes receipt status from acquisition basis.
UI-003	Transfer UI displays pending, accepted, declined, cancelled, expired and completed states.
UI-004	Reviewer UI displays deadline, remaining time, assignment, evidence, conflict declaration and required next action.
UI-005	Warnings appear 72 hours before current investigation/dispute/identifier-conflict deadlines.
UI-006	Authenticated users can submit later evidence and reconsideration requests.
UI-007	Core workflows support English and Amharic.
UI-008	Dispute and identifier conflict use the generic “Restricted pending review” public label.
UI-009	UI must not imply legal ownership verification or guaranteed device legitimacy.
UI-010	Core workflows work on mobile and desktop and minimize unnecessary network requests and large assets.
UI-011	Users can see case status and the next permitted action without exposing private reviewer notes.

15. Non-functional requirements

ID	Measurable requirement
NFR-001	Under 50 concurrent test users, 95% of ordinary API requests complete within 2 seconds, excluding file upload and email delivery.
NFR-002	Under the same conditions, 95% of public lookups complete within 2 seconds.
NFR-003	Target 99% availability during scheduled academic demonstration windows, excluding announced maintenance; this is a demo target, not a production SLA.
NFR-004	Critical state changes are transactional and auditable.
NFR-005	Core workflows support English and Amharic, including validation and error messages.
NFR-006	The app minimizes unnecessary network requests and large assets for constrained connectivity.
NFR-007	All mandatory security, permission, deadline, upload, privacy and concurrency tests pass before release.
NFR-008	Error responses never expose secrets, stack traces, private evidence or internal database details.
NFR-009	Timestamps are stored in UTC; UI displays localized times with the timezone clearly identified.
NFR-010	Retention schedules are configurable and enforceable; indefinite personal-data retention is prohibited by default.

16. Acceptance tests and traceability

Test	Traceability	Pass condition
AT-001	BR-004, FR-TRN-001	Eligible receipt-backed claim without blockers can initiate transfer.
AT-002	BR-004, FR-TRN-008	A recipient who acquired through AMANA can later request an eligible resale.
AT-003	BR-005	No eligible acquisition basis blocks AMANA-approved transfer but permits release if no other hold.
AT-004	BR-008, FR-INV-003	New investigation deadline is exactly 720 hours after UTC opening timestamp.
AT-005	BR-009, FR-INV-006	Only one extension of at most 360 hours is approved and audited.
AT-006	BR-010, FR-INV-007	At 72 hours before deadline, warning is generated; at 1,080 hours without outcome, escalation assigns a named authority.
AT-007	BR-011, FR-INV-004	Provisional restriction is triaged within 72 hours; overdue triage escalates.
AT-008	BR-012, FR-INV-001	Duplicate report links to existing active case and does not reset deadline.
AT-009	BR-013, FR-INV-011	Closed case cannot reopen without accepted materially new information.
AT-010	BR-016	Owner reporting own device retains case access; deactivation hold applies only if they are affected claimant.
AT-011	BR-020, FR-DEV-006	New identifier conflict is held without automatically freezing existing claimant.
AT-012	BR-019, FR-DSP-002	Current claimant can release association during dispute; independent restriction remains.
AT-013	BR-018, FR-TRN-004	Unaccepted transfer expires after 72 hours and releases only its own hold.
AT-014	BR-025, FR-TRN-002	Transfer cannot complete before explicit recipient acceptance.
AT-015	BR-021, FR-ACC-005	Deactivation is blocked for each listed hold and allowed when none applies.
AT-016	BR-023	Restricted association remains recorded after deactivation and has an accountable resolution path.
AT-017	FR-DEV-002	Invalid IMEI length, nonnumeric characters and failed Luhn checksum are rejected.
AT-018	SEC-006, SEC-007	Unsupported files, files over 10 MB, over-five-file submissions and signature/MIME mismatches are rejected.
AT-019	SEC-002	Direct API requests cannot bypass the permission matrix or object-level authorization.
AT-020	FR-LKP-004, FR-LKP-005	Status precedence is correct and dispute/identifier details remain generic.
AT-021	SEC-008, SEC-009	Each configured rate limit is enforced and does not automatically label the user malicious.
AT-022	BR-028, SEC-014	Reviewer with system-detectable or self-declared conflict cannot decide; recusal is audited.
AT-023	FR-EVD-003	Receipt review completes within five days or escalates; pending is never accepted implicitly.
AT-024	BR-014	Reconsideration is within 30 days and decided by an independent senior reviewer within 10 days.
AT-025	Section 6.5	Unsubstantiated/invalid outcome removes only its own provisional restriction.
AT-026	Section 5.2	Every permission-matrix row is tested for transfer, release, deactivation, new claim and public label.
AT-027	FR-INV-002	Report, case link/open, restriction and audit event commit or roll back atomically.
AT-028	FR-TRN-007, Section 12.3	Concurrent transfer and restriction requests cannot produce an unauthorized completed transfer.
AT-029	BR-010	At the 1,080-hour cap, escalation occurs without automatic theft finding; next review is recorded.
AT-030	FR-INV-012	Reopening records reason, new opening timestamp and deadline without erasing prior decision.
AT-031	Section 10	Declined, cancelled or expired transfer leaves source association unchanged and releases transfer-only hold.
AT-032	Section 8	Dispute outcome is recorded; voluntary release ends association dispute but not independent restrictions.

Test	Traceability	Pass condition
AT-033	UI-005	Reviewer warning appears 72 hours before each relevant current deadline.
AT-034	Section 14.1	Required in-app/email notifications are generated without private evidence leakage.
AT-035	UI-007	Core workflows render correctly in English and Amharic.
AT-036	BR-026, FR-LKP-002	Public lookup cannot reveal whether a named person registered or claimed a device.
AT-037	BR-024	Restricted device with no claimant can receive a restricted claim but cannot transfer or release.
AT-038	Section 6.6	Reconsideration follows its window/limit and does not erase prior decisions.
AT-039	FR-DEV-011, Section 12.2	Database prevents two active associations for one device under concurrent requests.
AT-040	Section 12.2	Database prevents two active theft investigations for one device under concurrent requests.
AT-041	Section 2.2	Effective public status is derived, not independently editable.
AT-042	FR-DSP-004, Section 8	Dispute warning, deadline, extension cap and escalation work as specified.
AT-043	Section 7	Identifier-conflict 15-day deadline, one 7-day extension and 22-day cap are enforced.
AT-044	SEC-018	Enumeration tests show no person-registration disclosure and rate limits are applied.
AT-045	SEC-019, SEC-020	Unauthorized evidence access is denied and sensitive data is absent from security logs.
AT-046	Section 17	A clean environment can be configured, migrated, seeded with synthetic data and tested using the README.
AT-047	Section 18	Retention jobs follow the approved schedule and do not delete records under active legal/approved hold.
AT-048	NFR-001, NFR-002	Load test with 50 concurrent users verifies 95th-percentile API and lookup targets.

16.1 Release-blocking criteria

Release is blocked if any mandatory acceptance test fails; a protected API bypasses authorization; a transfer completes under a blocker; a second active association or active theft case can be committed; public lookup reveals personal registration or case details; private evidence is publicly accessible; audit history can be silently altered by an ordinary user; or a continuing restriction lacks a source, accountable authority and review date.

17. Deployment and operations

• README documents supported Node.js, npm, PostgreSQL and browser versions and required environment variables using placeholders only.

• Database setup, migrations, seed data, backup and restore procedures are documented.

• First administrator/reviewer creation uses a secure bootstrap procedure; default passwords are prohibited.

• Frontend and backend startup, test, lint and build commands are documented.

• Private evidence storage and email configuration are documented.

• Production uses HTTPS, least-privilege database credentials, secure cookie configuration, protected backups and documented dependency updates.

• Secrets are supplied through environment/secret management and never committed to source control.

• Demonstration data uses synthetic users, device identifiers, receipts and reports, not real personal records.

• Operational monitoring covers failed login/recovery patterns, rate limits, overdue cases, failed notification delivery, and audit-protection failures.

18. Retention and unresolved approvals

Retention periods must be approved by the supervisor and checked against applicable Ethiopian legal and privacy obligations before production use. The system must support configurable retention and deletion/anonymization where permitted. A case hold must be documented, scoped, authorized and periodically reviewed; “retain everything forever” is not an acceptable default.

Data category	Approval required before production
Device and identifier history	[SUPERVISOR APPROVAL REQUIRED: period and lawful basis]
Claims and association history	[SUPERVISOR APPROVAL REQUIRED: period and lawful basis]
Closed reports and investigation records	[SUPERVISOR APPROVAL REQUIRED: period and lawful basis]
Evidence files	[SUPERVISOR APPROVAL REQUIRED: period, access and deletion method]
Minimal contact data after deactivation	[SUPERVISOR APPROVAL REQUIRED: period and access controls]
Security and audit events	[SUPERVISOR APPROVAL REQUIRED: period and access controls]
Designated project case authority	[SUPERVISOR APPROVAL REQUIRED: role/person and escalation route]
Inconclusive restriction review policy	[SUPERVISOR APPROVAL REQUIRED: criteria and appeal/release authority]

18.1 Privacy and retention principles

- Retain only data necessary for a defined operational, security, dispute or legal purpose.
- Public lookup never exposes prior claimants or personal history.
- Historical records may be retained while personal fields are anonymized when the approved schedule and legal basis permit.
- Account deactivation is not equivalent to immediate deletion of all records.
- All exceptional retention holds must have an authority, reason, scope and review date.

19. Definition of done

- All mandatory acceptance tests pass and results are recorded.
- Backend authorization and the permission matrix are enforced on every relevant route.
- Database constraints prevent duplicate active associations and active investigations.
- Report screening and 72-hour triage are implemented.
- Investigation, identifier-conflict, dispute, transfer and reconsideration deadlines are enforced.
- Every active restriction has a documented source, accountable owner and next review date.
- Public lookup passes privacy, enumeration and rate-limit tests.
- Upload validation, authentication, authorization, CSRF, XSS, audit and concurrency controls pass.
- English and Amharic core workflows are tested.
- Setup, deployment, backup, restore and testing instructions are complete.
- Retention schedule and designated case authority are approved before production use.
- Synthetic data is used for demonstrations; no unsupported claim of legal ownership verification is made.

20. Approval block and change log

Approval field	Entry
Project supervisor	_____
Student / project team	_____
Review date	_____
Approved version	_____
Approved exceptions / unresolved items	_____

20.1 Version 4.0 change log

- Consolidates the formal structure and requirement traceability of the PDF baseline with the detailed rules, matrix, workflows and minimum data fields from the pasted SRS.
- Retains the 30-day (720-hour) initial investigation deadline, one 360-hour extension and 1,080-hour cap; clearly records that this differs from the earlier 10-day proposal.
- Defines provisional triage, case outcomes, reconsideration, reopening, reviewer independence, and continuing-restriction accountability.
- Defines identifier-conflict and claim-dispute deadlines, transfer acceptance/expiry, later receipt evidence, and acquisition basis separately from receipt status.
- Adds 48 acceptance tests with explicit pass conditions and release-blocking criteria.
- Retains supervisor approval gates for retention periods, the designated case authority and inconclusive-restriction policy.